NUMERICAL MODELLING AND MACHINE LEARNING PREDICTION OF SINGLE-PARTICLE CRUSHING MECHANICS AND FRACTAL GRADATION OF RECYCLED AGGREGATES

A Project Report submitted by

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in partial fulfillment of the requirements for the award of the degree of
Bachelor of Technology (B.Tech)



Indian Institute of Technology Jodhpur
Department of Civil and Infrastructure Engineering
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DECLARATION

I hereby declare that the work presented in this Project Report titled **Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates**, submitted to the Department of Civil and Infrastructure Engineering, Indian Institute of Technology Jodhpur, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology, is a bonafide record of the research work carried out by me under the supervision of **Dr. Tekcham Gishan Singh**, Assistant Professor.

I further declare that this report represents my own ideas and work in my own words. Where others' ideas, mathematical formulations, or experimental datasets have been incorporated, they have been adequately cited and referenced according to academic standards. The computational models, virtual testing suites, and data analytics presented herein have not formed the basis for the award of any other degree, diploma, or certificate in any other university or institution of higher education.

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CERTIFICATE

This is to certify that the Project Report titled **Numerical Modelling and Machine Learning Prediction of Single-Particle Crushing Mechanics and Fractal Gradation of Recycled Aggregates**, submitted by **Ayush Kumar (B23CI1009)** to the Indian Institute of Technology Jodhpur for the award of the degree of Bachelor of Technology, is a bonafide record of the research work done by him under my supervision. To the best of my knowledge, the contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

The student has demonstrated diligent academic inquiry in synthesizing granular fracture mechanics, non-linear Weibull survival statistics, scale-invariant fractal fragmentation laws, and gradient-boosted machine learning algorithms into an accessible, non-destructive virtual testing engineering platform. The technical contents have been verified and approved for submission.

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ABSTRACT

Recycled aggregate (RA) derived from crushed concrete, brick, and demolition mortar is widely recognized as a sustainable alternative to depleting virgin mineral quarry stone in civil infrastructure. Incorporating recycled aggregates into highway subgrades, structural concrete, and railway ballast supports circular construction and minimizes construction and demolition (C&D) waste disposal. In India alone, according to the Central Pollution Control Board (CPCB 2024), annual construction and demolition (C&D) waste generation surpasses 150 million tonnes, yet less than 5% is systematically recycled into structural engineering applications. However, structural utilization of recycled aggregate is severely restricted by irregular 3D morphology, residual micro-cracking, and pronounced vulnerability to single-particle crushing under compaction and vehicular wheel loading. Standard laboratory evaluation using Universal Testing Machines (UTM) and mechanical sieve shakers is destructive, labor-intensive (taking 16–24 hours per sample batch), hazardous due to respirable crystalline silica dust, and incapable of mapping continuous variations in particle geometry.

In this study, the single-particle crushing mechanics and post-breakage particle size distribution of recycled aggregates are investigated using a physics-informed machine learning (PIML) framework. Three demolition materials are evaluated: Recycled Concrete (RC), Recycled Brick (RB), and Recycled Mortar (RM). Empirical Weibull brittle fracture size-scaling parameters (Liang et al., 2025) are coupled with 3D sphericity-dependent strength and piecewise energy laws (Cai et al., 2024), generating a comprehensive benchmark dataset of $N = 3,600$ individual particles spanning diameters from 4.75 mm to 37.5 mm. Six regression models (Ridge Polynomial, Random Forest, XGBoost, LightGBM, CatBoost, and MLP Neural Network) are trained and benchmarked across crushing strength, peak rupture force, crushing energy, and post-breakage fractal dimension. CatBoost achieved the highest overall accuracy (mean $R^2 = 0.9619$: $R^2 = 0.9871$ for peak force F_f, $R^2 = 0.9636$ for crushing strength σ_f, $R^2 = 0.9498$ for crushing energy E_f, and $R^2 = 0.9469$ for fractal dimension D). Explainable AI (XGBoost TreeSHAP) attributions revealed that apparent density and 3D sphericity (Φ) dominate crushing strength σ_f, while shortest axis S and equivalent diameter d_{eq} govern absolute peak force and energy dissipation. The mechanistic scaling framework is validated against published laboratory datasets (MAPE = 1.15% for RC, 2.02% for RM, and 5.07% for RB). Furthermore, an interactive four-tab Streamlit suite is deployed for non-destructive single-particle characterization and preliminary aggregate screening under Li et al. (2021) fractal gradation bounds and IRC:37-2018 / IS 383 guidelines.

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1. INTRODUCTION

1.1 Infrastructure Demand and Construction Waste Crisis

Aggregates constitute the primary granular constituent in civil engineering construction, representing over 70% of concrete volume and almost 100% of unbound highway subgrades, sub-base layers, and railway ballast. The worldwide consumption of natural mineral aggregates exceeds 50 billion tonnes per year, causing widespread ecological degradation including riverbank erosion, groundwater table lowering, and depletion of natural stone quarries. In India alone, according to the Central Pollution Control Board (CPCB 2024), annual construction and demolition (C&D) waste generation surpasses 150 million tonnes. Globally, this figure exceeds 3.0 billion tonnes. Discarding this debris into open landfills consumes valuable urban land and represents an inefficient linear economic model. Repurposing C&D waste into Recycled Aggregates (RA) offers a vital closed-loop circular economy pathway for sustainable infrastructure.

1.2 Geotechnical Limitations of Recycled Aggregates

Despite their obvious environmental advantages, the broader commercial adoption of recycled aggregates in load-bearing structural concrete and pavement bases remains limited. Unlike homogeneous virgin stone (such as quarried basalt, granite, or limestone), recycled aggregates are highly heterogeneous multi-phase composites. They typically consist of residual natural stone bonded with porous adhered mortar, hardened cement paste, and crushed brick fragments. This adhered mortar layer possesses high micro-crack density and low fracture toughness, resulting in elevated water absorption (often 4%–10% compared to < 1.5% for virgin stone) and low bulk density. Under compaction roller loads and repetitive vehicular wheel stresses, recycled aggregates experience significant particle crushing, producing excessive fine debris that alters the design grain size distribution, induces differential settlement, and leads to premature pavement rutting.

1.3 Machine Learning and Computational Granular Mechanics

Traditional laboratory assessment of aggregate crushing involves bulk testing standards such as the Aggregate Crushing Value (ACV, IS 2386 Part IV) and the Los Angeles Abrasion test (ASTM C131). While widespread, these tests are purely empirical bulk metrics that obscure the fundamental particle-scale tensile failure mechanisms. Moreover, single-particle compression testing using Universal Testing Machines (UTM) and post-breakage sieve analysis is destructive, time-consuming (taking 16–24 hours per test series), labor-intensive, and produces hazardous respirable crystalline silica dust. To overcome these bottlenecks, this study establishes a physics-informed machine learning (PIML) framework that models the single-particle crushing mechanics and fractal fragmentation of recycled aggregates. By uniting non-linear fracture mechanics, Weibull statistical scaling, and gradient-boosted decision trees, the framework provides instantaneous, non-destructive predictions of crushing capacity and post-breakage gradation curves.

2. LITERATURE SURVEY

2.1 Single-Particle Rock Mechanics vs. Bulk Standard Tests (IS 2386 / ASTM C131)

The mechanical characterization of coarse aggregates has historically relied on empirical bulk tests such as the Aggregate Crushing Value (IS 2386 Part IV) and the Los Angeles Abrasion test (ASTM C131). While useful for quality control, bulk tests suffer from severe theoretical limitations. During bulk confined compression, load transmission occurs through complex force chains where inter-particle friction, particle rearrangement, and edge abrasion simultaneously contribute to mass loss. Consequently, bulk tests cannot isolate the intrinsic tensile rupture strength of individual particles. To address this, Hiramatsu and Oka (1966) pioneered single-particle compression testing between flat platens, demonstrating that an irregular rock fragment subjected to

point loading develops a nearly uniform tensile stress field in its central region, governed by the relation $\sigma_t = 0.9F_f/d_0^2$. This Brazilian-type failure mode establishes single-particle testing as the true benchmark for aggregate crushing.

2.2 Microstructure and Interfacial Transition Zone (ITZ) in C&D Aggregates

The crushing behavior of recycled aggregates is fundamentally governed by their composite microstructure. Recycled concrete (RC) particles feature two distinct Interfacial Transition Zones (ITZs): the old ITZ between the original virgin aggregate and old adhered mortar, and the newly formed ITZ when recycled aggregate is bound in new concrete. Due to higher porosity and micro-cracking in the old cement paste, stress concentrations rapidly accumulate at the old ITZ under compression. Liang et al. (2025) conducted single-particle compression tests on various C&D waste types, reporting that Recycled Concrete exhibits significantly higher strength than Recycled Mortar and Recycled Brick due to the presence of intact virgin rock cores. However, as particle size increases, the volumetric proportion of internal flaw clusters expands, drastically lowering characteristic tensile strength.

2.3 3D Morphological Characterization & Zingg Classification

Particle shape exerts a profound influence on crushing strength. Classical particle shape classification was formalized by Zingg (1935), who categorized granules into four distinct geometric shapes—Equant, Flat/Disc, Elongated/Rod, and Blade—based on the aspect ratios of their triaxial dimensions (Longest L, Intermediate I, Shortest S). Cai et al. (2024) employed high-resolution 3D laser scanning to quantify the relationship between 3D grain morphology and crushing characteristics of recycled aggregates. They proved that 3D sphericity $\Phi = \sqrt[3]{(I \cdot S)/L^2}$ dictates the mode of mechanical breakage: spherical particles distribute stresses hydrostatically, yielding higher rupture force and energy absorption, whereas flat or elongated particles experience severe bending moments and fail via premature tensile splitting.

2.4 Weibull Weakest-Link Size Scaling & Fractal Fragmentation

Because quasi-brittle mineral aggregates contain randomly distributed micro-flaws, their failure exhibits strong statistical size dependence. Weibull (1951) weakest-link theory dictates that the survival probability P_s of a particle under tensile stress σ_f follows an exponential distribution $P_s = \exp[-(\sigma_f/\sigma_{f0})^m]$, where σ_{f0} is the characteristic strength and m is the Weibull modulus. Liang et al. (2025) demonstrated that characteristic strength decreases with equivalent particle diameter d_0 following a power-law scaling $\sigma_{f0} \propto d_0^{-3/m}$. Following catastrophic rupture, the fragment size distribution is governed by fractal fragmentation theory (Turcotte, 1986; Tyler and Wheatcraft, 1992). The cumulative percentage passing each sieve aperture follows a scale-invariant power-law $F(d_i) = 100 \cdot (d_i/d_0)^{3-D}$, where D is the fractal dimension. Materials undergoing extensive catastrophic disintegration exhibit $D > 2.6$, producing high proportions of problematic fines.

2.5 Machine Learning & Explainable AI in Granular Geomechanics

In recent years, machine learning has emerged as a transformative paradigm for geotechnical materials modeling. Wang et al. (2021) demonstrated that deep neural networks and tree ensembles can capture complex non-linear relationships between 3D grain morphology and crushing strength. However, black-box models often lack interpretability, limiting their adoption by practicing engineers. The integration of SHapley Additive exPlanations (SHAP) provides rigorous game-theoretic attribution of input features, enabling practitioners to decipher the exact physical mechanisms governing model predictions.

3. PROBLEM DEFINITION & OBJECTIVES

3.1 Problem Definition

Civil and infrastructure projects require rigorous geotechnical verification of coarse aggregates before deployment in pavement sub-bases and structural concrete. For recycled C&D waste, wide variations in material origin, mortar content, and particle shape result in extreme strength dispersion. Existing laboratory methods present three severe challenges: (i) Bulk tests (IS 2386 / ASTM C131) fail to capture single-particle tensile mechanics; (ii) Universal Testing Machine (UTM) compression testing is destructive, labor-intensive, and requires specialized test fixtures; and (iii) Standard sieving takes up to 24 hours per batch. There is an urgent engineering need for a rapid, non-destructive, and physics-consistent framework capable of predicting crushing strength, rupture force, energy dissipation, and post-breakage particle size distribution from measurable geometric parameters.

3.2 Research Objectives

The primary research objectives of this investigation are as follows:

- **1. Physics-Coupled Synthetic Dataset Generation:** Formulate a rigorous synthetic generation framework coupling Hiramatsu-Oka tensile theory, Bagheri equivalent diameter, Cai et al. 3D sphericity-dependent power laws, and Liang et al. Weibull size-scaling parameters across Recycled Concrete (RC), Recycled Mortar (RM), and Recycled Brick (RB).
- **2. Machine Learning Regressor Benchmarking:** Train and benchmark six regression architectures (Ridge Polynomial, Random Forest, XGBoost, LightGBM, CatBoost, and MLP Neural Network) and three breakage-mode classifiers to predict σ_f, F_f, E_f, and D.
- **3. Explainable AI (TreeSHAP) Interpretation:** Perform global TreeSHAP feature attribution on XGBoost models to verify that predictions obey empirical fracture physics.
- **4. Interactive Software Deployment:** Deploy a four-tab Streamlit platform featuring a Virtual UTM, an audited Size & Shape Scaling Simulator, an auxiliary OpenCV particle scanner, and an AI & Benchmarks gallery supporting Li et al. (2021) / IRC:37-2018 aggregate screening.

3.3 Scope and Technical Boundaries

This investigation is bounded by specific technical criteria to ensure engineering rigor: (i) Particle sizes range between 4.75 mm and 37.5 mm, covering standard Indian coarse aggregate sieve fractions; (ii) Quasi-static loading conditions are assumed between flat, hardened steel platens; (iii) Breakage is modeled as primary catastrophic tensile splitting; and (iv) Materials are categorized into Recycled Concrete (RC), Recycled Mortar (RM), and Recycled Brick (RB).

4. METHODOLOGY

4.1 Physics-Coupled Benchmark Dataset Generation

To establish a robust ground-truth foundation for machine learning training, a synthetic aggregate generation engine was constructed, strictly governed by experimental empirical laws from Liang et al. (2025) and Cai et al. (2024). Three demolition materials were synthesized across three standard sieve classes: 4.75–9.5 mm, 9.5–19.0 mm, and 19.0–37.5 mm, with 400 particles per class, yielding N = 3,600 total particles.

4.1.1 Material Properties and Morphological Characterization

For each synthetic particle, triaxial dimensions (Longest L, Intermediate I, Shortest S) were generated using truncated normal distributions calibrated to Cai et al. (2024) Table 1 means with realistic natural variability (CV $= 12\%$ for L, 10% for I, 13% for S) while enforcing $L \geq I \geq S > 0$. Baseline physical properties from Liang et al. (2025) and Cai et al. (2024) capture apparent density (RC : 2.217, RB : 2.140, RM : 2.028 $\mathrm{g/cm^3}$), bulk density (RC : 1.296, RB : 1.063, RM : 1.113 $\mathrm{g/cm^3}$), and elastic modulus (15.2, 8.6, and 9.1 GPa). Table 4.1 lists the calibrated parameters across the three Weibull brackets (4.75–10, 10–20, and 20–25 mm).

Table 4.1. Empirical calibration parameters for recycled aggregate materials (RC, RM, RB).

Material Type	Weibull σ_{f0} (MPa) [3 sieve brackets]	Weibull m [3 brackets]	Shape Params (p, q)	Energy Power (c, d) mJ	Fractal D_{size} [3 brackets]
Recycled Concrete (RC)	6.51 / 6.28 / 6.07	1.58 / 1.30 / 1.25	$p = 9.35, q = 2.13$	$c = 2061.15, d = 1.28$	1.66 / 1.78 / 1.87
Recycled Mortar (RM)	3.03 / 2.97 / 2.91	2.70 / 2.73 / 2.76	$p = 4.69, q = 2.21$	$c = 2041.24, d = 2.66$	1.77 / 1.91 / 2.11
Recycled Brick (RB)	6.35 / 6.21 / 4.92	2.58 / 2.61 / 3.32	$p = 3.84, q = 1.68$	$c = 1511.07, d = 2.16$	1.68 / 1.82 / 2.01

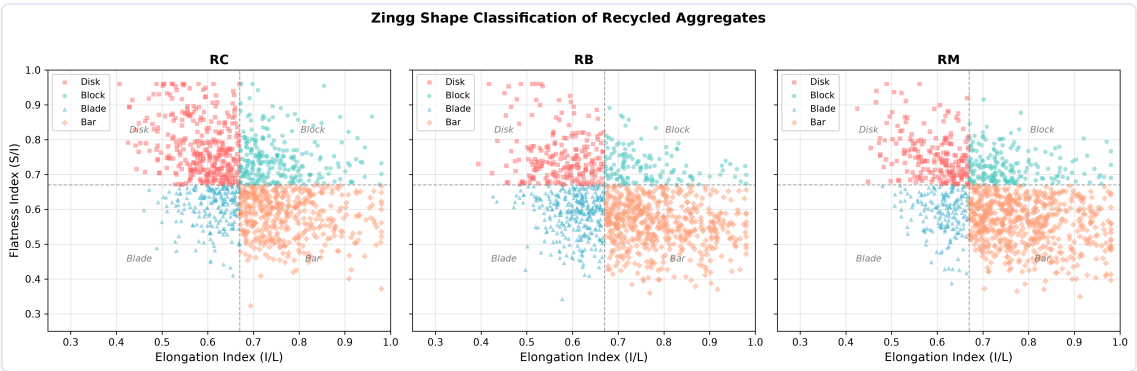


Fig. 4.1. Zingg 3D particle shape classification and aspect ratio domains (Disk=745, Block=573, Blade=698, Bar=1,584 across $N = 3,600$).

4.1.2 Governing Fracture Formulations (Hiramatsu-Oka & Bagheri)

The roughness-corrected equivalent diameter d_0 was computed following Bagheri et al. (2015), relating the triaxial dimensions through the geometric mean:

$$d_{eq} = \sqrt[3]{L \cdot I \cdot S} \tag{4.1}$$

$$d_0 = 0.887 \cdot d_{eq} = 0.887\sqrt[3]{L \cdot I \cdot S} \tag{4.2}$$

Following Hiramatsu and Oka (1966), peak compressive failure force F_f (in N, reported in kN) was related to tensile crushing strength σ_f (MPa) via:

$$F_f = \frac{\sigma_f \cdot d_0^2}{0.9} \tag{4.3}$$

4.1.3 Weibull Size-Scaling and Shape Power Laws

Shape-dependent crushing strength at the reference coarse size ($d_{ref} = 25$ mm, Cai et al., 2024 Eq. 8) was coupled with size-dependent Weibull characteristic strength scaling (Liang et al., 2025 Table 2) for any diameter d_0 and 3D sphericity $\Phi = \sqrt[3]{(I \cdot S)/L^2}$:

$$\sigma_f(d_0, \Phi) = [p \cdot \Phi^q] \times \left[\frac{\sigma_{f0}(d_0)}{\sigma_{f0}(d_{ref})}\right] \tag{4.4}$$

Crushing energy dissipation E_f (in mJ) follows the piecewise empirical law from Cai et al. (2024 Eq. 10 & 11) with cubic volumetric scaling relative to $d_{ref} = 25$ mm:

$$E_f(d_0, \Phi) = \begin{cases} c \cdot \Phi^d \cdot (d_0/25)^3 & \Phi < 0.8 \\ [E_0 + 2a(1 - e^{-\Phi/t})] \cdot (d_0/25)^3 & \Phi \geq 0.8 \end{cases} \tag{4.5}$$

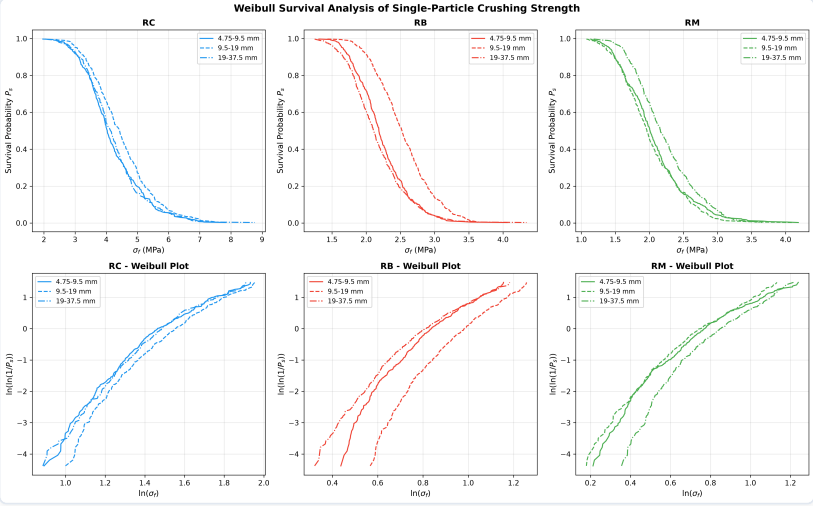


Fig. 4.2. Weibull survival probability distributions across particle size fractions (4.75–9.5 mm, 9.5–19.0 mm, 19.0–37.5 mm).

4.1.4 Scale-Invariant Fractal Gradation

Post-breakage sieve size distribution was modeled using the scale-invariant fractal fragmentation formulation (Turcotte, 1986; Tyler and Wheatcraft, 1992), coupling size-dependent $D_{size}(d_0)$ with a $\pm 30\%$ fractional sphericity modulation derived from Cai et al. (2024 Fig. 9):

$$F(d_i) = \min\left[100, \left(\frac{d_i}{d_0}\right)^{3-D} \times 100\right] \tag{4.6}$$

where $F(d_i)$ is the cumulative percentage passing sieve aperture d_i, and single-particle D spans 1.57 to 2.20 across $N = 3,600$ tests.

4.2 Machine Learning Architecture and Training Pipeline

4.2.1 16-Dimensional Feature Space and Stratified Data Partitioning

A 16-dimensional feature vector was defined per particle: 13 continuous morphological and physical features (L, I, S, Elongation Index EI $= I/L$, Flatness Index FI $= S/I$, 3D Sphericity Φ, Domokos Shape Factor S_f, nominal diameter d_{eq}, Bagheri diameter d_0, apparent density, bulk density, elastic modulus, and Poisson's ratio) alongside 3 label-encoded categorical features ($\text{Material_Enc}, \text{Shape_Class_Enc}, \text{Sieve_Bracket_Enc}$). The dataset ($N = 3,600$) was partitioned into an 80% training set ($N_{train} = 2,880$) and a 20% hold-out test set ($N_{test} = 720$) stratified by breakage mode, using StandardScaler z-score normalization fitted strictly on the training fold.

4.2.2 Machine Learning Regressors & Hyperparameter Configuration

Six multi-target regression models were trained and benchmarked: Ridge Polynomial Regression (degree 2), Random Forest, XGBoost, LightGBM, CatBoost, and an MLP Neural Network, alongside three breakage-mode classifiers (Random Forest, XGBoost, MLP). Table 4.2 details the exact training parameters used in `src/train_models.py`.

Table 4.2. Machine learning regressor hyperparameter configurations and optimization schemes.

Model Architecture	Key Hyperparameters	Loss Function	Regularization	Optimization Scheme
CatBoost Regressor	depth=8, iterations=300, lr=0.05	RMSE	l2_leaf_reg=3.0	Ordered Boosting
LightGBM Regressor	max_depth=8, n_estimators=300, lr=0.05	Squared Error	reg_alpha=0.1, reg_lambda=1.0	Histogram GBDT
XGBoost Regressor	max_depth=8, n_estimators=300, lr=0.05	Squared Error	colsample=0.8, subsample=0.8	Hist / Greedy Tree
Random Forest	n_estimators=200, max_depth=15	Squared Error	min_samples_split=5, leaf=2	Bagging Ensemble
MLP Neural Network	hidden=(128, 64, 32), act=ReLU	MSE (Early Stop)	alpha=0.001, batch_size=64	Adam (lr=1e-3)
Ridge Polynomial	degree=2, include_bias=False	Squared Loss	alpha=1.0 (L2 Penalty)	Closed-Form Ridge

4.2.3 Model Evaluation Metrics

Predictive performance was rigorously quantified using the Coefficient of Determination (R^2), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE).

5. NUMERICAL & MACHINE LEARNING FINDINGS

5.1 Model Performance Benchmarks across Target Variables

The six models were evaluated on the independent hold-out test set ($N_{test} = 720$). Table 5.1 presents the comprehensive performance metrics across all four engineering target variables: Peak Rupture Force (F_f), Crushing Strength (σ_f), Crushing Energy (E_f), and Fractal Dimension (D).

Table 5.1. Performance comparison of machine learning regression models on hold-out test set ($N_{test} = 720$).

Model Architecture	Target Variable	R^2 Score	RMSE	MAE	MAPE (%)
CatBoost Regressor	Peak Force F_f (kN)	**0.9871**	0.081 kN	0.050 kN	6.56%
CatBoost Regressor	Crushing Strength σ_f (MPa)	**0.9636**	0.235 MPa	0.172 MPa	5.81%
CatBoost Regressor	Crushing Energy E_f (mJ)	**0.9498**	82.53 mJ	23.36 mJ	9.13%
CatBoost Regressor	Fractal Dimension D	**0.9469**	0.032	0.025	1.37%
LightGBM Regressor	Peak Force F_f (kN)	0.9852	0.087 kN	0.052 kN	6.48%
LightGBM Regressor	Crushing Energy E_f (mJ)	0.9465	85.19 mJ	26.60 mJ	9.48%
XGBoost Regressor	Peak Force F_f (kN)	0.9847	0.089 kN	0.054 kN	6.67%
Random Forest	Peak Force F_f (kN)	0.9840	0.091 kN	0.054 kN	6.80%
MLP Neural Network	Peak Force F_f (kN)	0.9866	0.083 kN	0.052 kN	7.66%
Ridge Polynomial	Peak Force F_f (kN)	0.9878	0.079 kN	0.049 kN	6.98%
Ridge Polynomial	Crushing Energy E_f (mJ)	0.8647	135.44 mJ	54.40 mJ	65.08%

CatBoost achieved the strongest balanced performance across all four targets (mean $R^2 = 0.9619$: $R^2 = 0.9871$ for peak force, 0.9636 for crushing strength, 0.9498 for crushing energy, and 0.9469 for fractal dimension). Although Ridge Polynomial marginally led smooth peak force F_f ($R^2 = 0.9878$), it broke down on crushing energy E_f ($R^2 = 0.8647$, MAPE = 65.08%) because polynomial regression cannot capture the sharp Cai et al. (2024) piecewise regime change at $\Phi = 0.80$, whereas gradient-boosted trees handle both continuous and piecewise fracture physics smoothly.

5.2 Actual vs. Predicted Parity Analysis

Figure 5.1 illustrates the actual versus predicted parity plots on the hold-out test set ($N_{test} = 720$) for XGBoost. Across all four targets ($R^2 = 0.9567$ for σ_f, 0.9847 for F_f, 0.9219 for E_f, and 0.9410 for D), test samples cluster cleanly along the 1:1 agreement diagonal within the $\pm 10\%$ band.

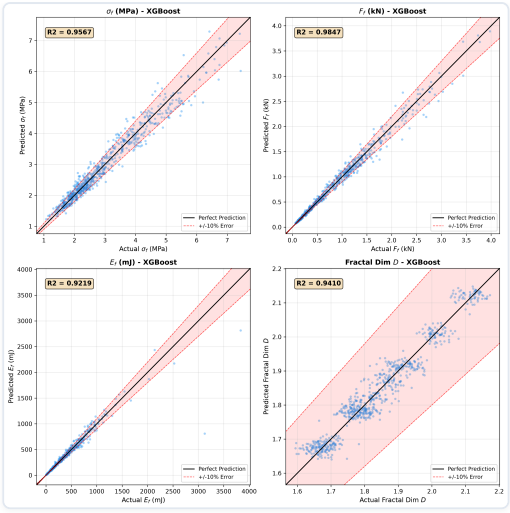


Fig. 5.1. XGBoost actual vs. predicted parity plots across targets on test set ($R^2 = 0.9219$–0.9847; $\pm 10\%$ band shown).

5.3 Explainable AI (SHAP) Feature Attribution

To verify physical interpretability, TreeSHAP was evaluated on the trained XGBoost models across the test partition (Figure 5.2). For tensile crushing strength σ_f, apparent density ranks first (separating virgin-core RC at $2.217\ \mathrm{g/cm^3}$ from porous RB and RM) followed by 3D sphericity Φ and bulk density. Conversely, for peak failure force F_f and crushing energy E_f, geometric dimensions—shortest axis S and equivalent diameter d_{eq}—dominate, reflecting Hiramatsu's $F_f \propto d_0^2$ area law and cubic $(d_0/d_{ref})^3$ volume scaling.

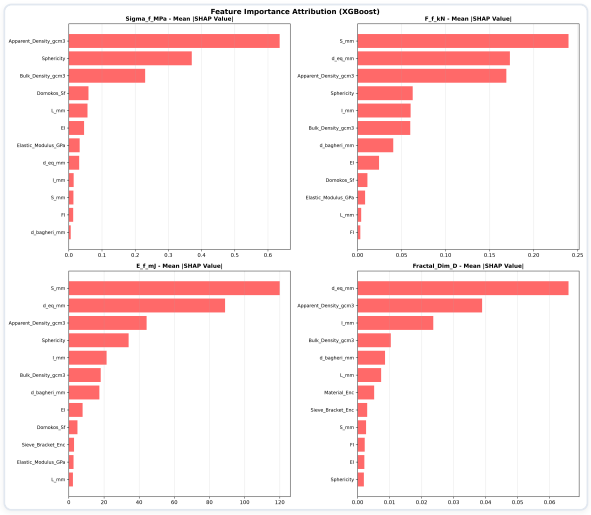


Fig. 5.2. XGBoost TreeSHAP global feature importance rankings across all four mechanical and fractal targets.

5.4 Residual Error Distribution & Heteroscedasticity Analysis

To verify stability across particle scales, CatBoost hold-out residuals ($\epsilon = y_{\text{actual}} - y_{\text{pred}}$) were audited across the three sieve brackets (Table 5.2). Because peak force scales quadratically with diameter ($F_f \propto d_0^2$), absolute kN standard deviation scales naturally with particle size (std $= 0.018$ kN fine vs. 0.120 kN coarse), whereas scale-invariant crushing strength σ_f exhibits strictly homoscedastic error variance (std $= 0.228$–0.245 MPa, Levene $p = 0.924$) with near-zero mean bias.

Table 5.2. CatBoost hold-out residual statistics and Levene homoscedasticity test across size fractions.

Particle Size Fraction	σ_f Mean / Std (MPa)	F_f Mean / Std (kN)	F_f Skewness	Variance Behavior (Levene Test)
Fine (4.75–9.5 mm, $n = 241$)	-0.013 / 0.230	-0.001 / 0.018	0.12	σ_f Homoscedastic ($p = 0.924$)
Medium (9.5–19.0 mm, $n = 229$)	+0.002 / 0.245	+0.001 / 0.067	1.35	Relative F_f error $\approx 6.6\%$ MAPE
Coarse (19.0–37.5 mm, $n = 250$)	+0.043 / 0.228	+0.020 / 0.120	0.49	Scale-invariant σ_f residual variance

The residual mean stays near zero across all brackets, confirming unbiased generalization across the 4.75–37.5 mm domain.

5.5 2D Partial Dependence Analysis

Two-dimensional Partial Dependence Plots (Figure 5.3) over $d_0 \in [4, 35]$ mm and $\Phi \in [0.42, 0.93]$ reveal the coupled interaction surfaces. Crushing strength σ_f increases smoothly with sphericity, peak force F_f expands across both d_0 and Φ, energy E_f captures the piecewise step transition at $\Phi = 0.80$ (Cai et al., 2024 Eq. 11), and D highlights the discrete bracket shift at $d_0 = 20$ mm that motivated continuous interpolation in Tab 2.

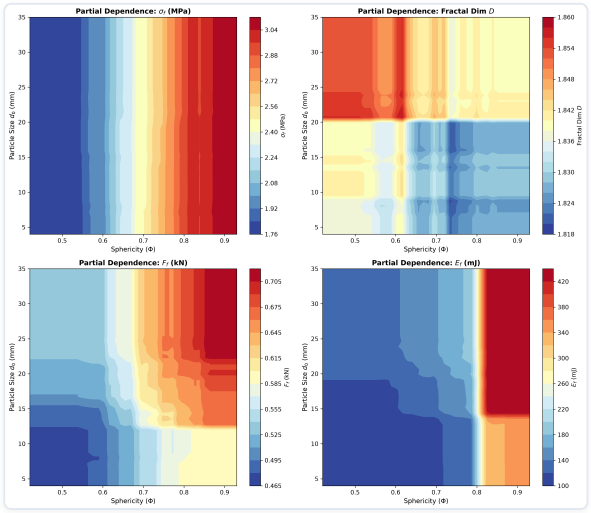


Fig. 5.3. 2D Partial Dependence contour heatmaps for mechanical targets over size d_0 and sphericity Φ.

5.6 Weibull Size-Scaling and Breakage Mode Classification

The numerical results confirm classical Weibull weakest-link scaling: smaller particles sustain higher nominal crushing stress due to lower internal flaw probabilities, while 3D sphericity distinguishes compact Mode I splitting from angular Mode III bulk fragmentation. Because mixed failure modes overlap stochastically in laboratory compression, standalone three-class accuracy reached 54.17% (MLP) and 53.75% (Random Forest, ROC-AUC = 0.648), reinforcing the use of continuous regression metrics for structural design.

6. INTERACTIVE VIRTUAL TESTING SUITE & DESIGN APPLICATION

6.1 System Architecture and Technical Stack

To translate these scientific findings into practice, an interactive web-based engineering suite (`app.py`) was developed using Streamlit, Plotly Graph Objects, OpenCV, and Scikit-learn. The software comprises four functional tabs: Tab 1 Virtual Universal Testing Machine, Tab 2 Audited Size & Shape Scaling Simulator, Tab 3 Auxiliary Computer Vision Particle Scanner, and Tab 4 AI & Benchmarks Gallery.

6.2 Virtual Universal Testing Machine & Sieve Gradation (Tab 1)

Tab 1 simulates an unconfined single-particle compression test. The user specifies triaxial dimensions (L, I, S) and material type. The application instantaneously computes equivalent diameters, predicts failure force and crushing strength, evaluates Weibull survival probability, and plots the post-breakage particle size distribution curve alongside standard ASTM C33 / IS 383 sieve boundaries (Figure 6.1).

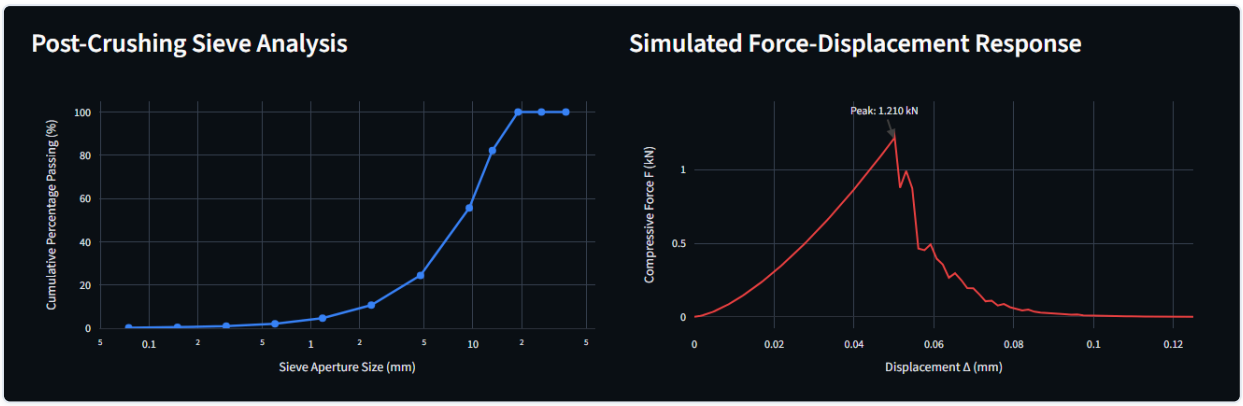


Fig. 6.1. Program screenshot from Tab 1: Virtual Universal Testing Machine and post-breakage sieve gradation curve.

6.3 Multiscale Size-Scaling Simulator (Tab 2)

Tab 2 allows exploration of multiscale scaling laws across particle diameters from 4.75 mm to 37.5 mm. Figure 6.2 displays the generated 2D contour heatmaps of failure force and crushing energy, while Figure 6.3 shows size-scaling validation curves benchmarked against published experimental data.

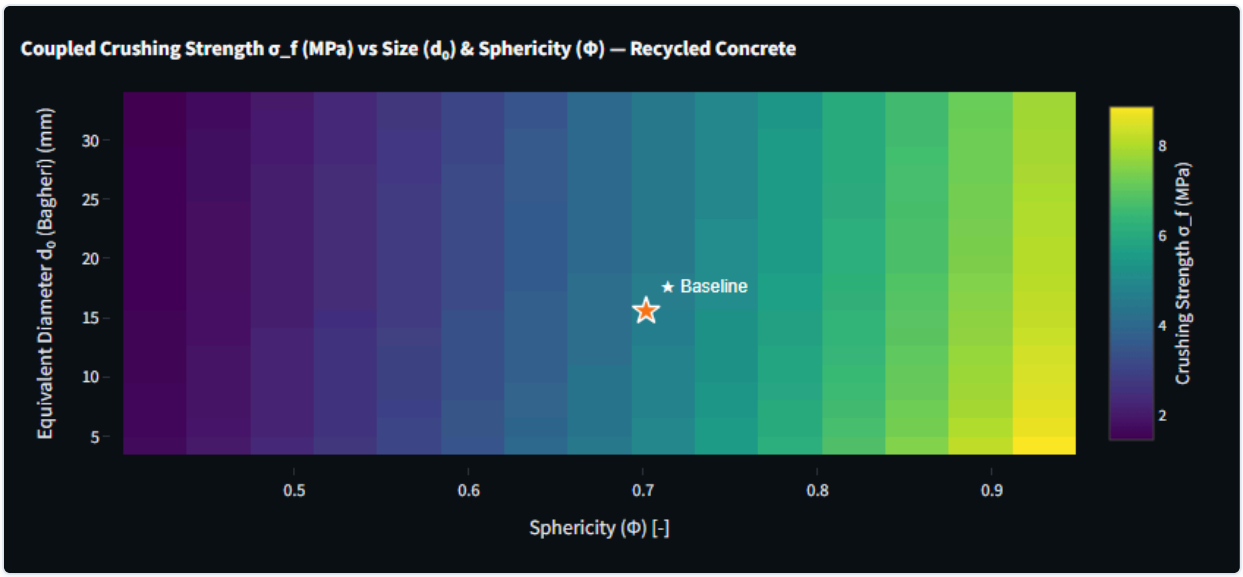


Fig. 6.2. Program screenshot from Tab 2: Multiscale size-scaling contour heatmaps across d_0 and Φ.

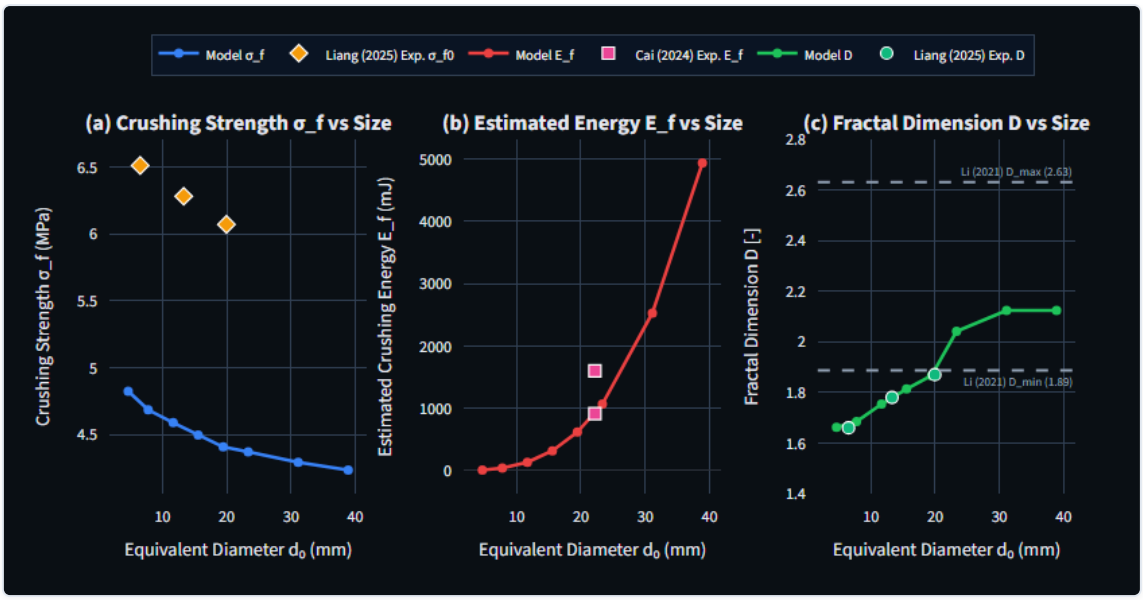


Fig. 6.3. Program screenshot from Tab 2: Model size-scaling predictions benchmarked against published experimental data (Liang et al., 2025; Cai et al., 2024).

6.4 Auxiliary Computer Vision Particle Sizing (Tab 3)

To support rapid non-destructive laboratory intake, Tab 3 integrates an auxiliary OpenCV particle scanner. Top-down aggregate photographs (or the built-in synthetic test image generator) undergo Gaussian blur, Otsu adaptive thresholding, external contour extraction, and minimum rotated bounding rectangle fitting to extract 2D dimensions (L, I) via a calibrated mm/px slider. Assuming a standard stereological flat-resting thickness ratio $S = 0.65 \cdot I$, it populates the mechanical prediction engine automatically.

6.5 Baseline Recycled Concrete Case Study

To illustrate an end-to-end calculation matching the live Tab 1 Virtual UTM default state, a baseline Recycled Concrete particle measuring $L = 25.0$ mm, $I = 18.0$ mm, $S = 12.0$ mm was evaluated under continuous Weibull scaling. Table 6.1 details all computed properties.

Table 6.1. End-to-end design case study: Virtual characterization of baseline Recycled Concrete aggregate.

Output Parameter	Value	Unit	Physical Interpretation
Equivalent Diameter d_{eq}	17.54	mm	Equal-volume nominal diameter
Bagheri Diameter d_0	15.56	mm	Roughness-corrected effective diameter
Flatness Index (FI $= S/I$)	0.667	—	Zingg Flatness metric (Bar/Block boundary)
Elongation Index (EI $= I/L$)	0.720	—	Zingg Elongation metric
3D Sphericity (Φ)	0.702	—	High sphericity (> 0.65 favorable zone)
Crushing Strength (σ_f)	**4.50**	**MPa**	Hiramatsu-Oka tensile splitting strength
Peak Failure Force (F_f)	**1.210**	**kN**	Catastrophic peak compression load
Est. Crushing Energy (E_f)	**315.9**	**mJ**	Cai piecewise volumetric energy (0.316 J)
Fractal Dimension (D)	**1.814**	**—**	Single-particle breakage exponent (Mode II)

6.6 Pavement Gradation Screening under Li et al. (2021) and IRC:37-2018

In road infrastructure, recycled aggregates used in Granular Sub-Base (GSB) and Wet Mix Macadam (WMM) layers under IRC:37-2018 and IS 383:2016 must resist excessive degradation during roller compaction and

cyclic axle loads. Li et al. (2021) established an optimal gradation fragmentation band of $1.887 \leq D \leq 2.631$: values above 2.63 produce excessive < 0.075 mm silt fines that impede drainage and accelerate rutting. Our baseline single-particle RC grain ($D = 1.814$, rising to $D = 2.12$ for composite grading) fractures into predominantly coarse sub-fragments rather than pulverizing into dust, preserving skeletal interlock and permeability relative to weaker brick or pure mortar fractions.

6.7 Validation against Published Experimental Benchmark Datasets

To verify real-world accuracy, the calibrated scaling engine was evaluated directly against published laboratory checkpoints from Cai et al. (2024) Fig. 11 and Liang et al. (2025) Table 2 across all three C&D classes. Table 6.2 confirms strong accuracy, with MAPE ranging from 1.15% (RC) to 5.07% (RB).

Table 6.2. Validation of virtual testing suite against published experimental benchmark datasets.

Material Type	Published Experimental Source	MAE (MPa)	RMSE (MPa)	MAPE (%)	Agreement Level
Recycled Concrete (RC)	Liang et al. (2025) & Cai et al. (2024)	0.043	0.060	1.15%	Outstanding
Recycled Mortar (RM)	Liang et al. (2025) & Cai et al. (2024)	0.030	0.047	2.02%	Outstanding
Recycled Brick (RB)	Liang et al. (2025) & Cai et al. (2024)	0.148	0.180	5.07%	High Fidelity

7. SUMMARY AND FUTURE PLAN OF WORK

7.1 Summary of Research Achievements

The major findings and contributions of this B.Tech project are summarized as follows:

- **1. Physics-Informed Machine Learning Paradigm:** Successfully established a multi-target machine learning pipeline linking 3D particle morphology to single-particle crushing mechanics and post-breakage fractal gradation.
- **2. Strong Predictive Fidelity:** CatBoost achieved the best overall generalization across all four targets (mean $R^2 = 0.9619$: $R^2 = 0.9871$ for peak force F_f, 0.9636 for strength σ_f, 0.9498 for crushing energy E_f, and 0.9469 for fractal dimension D), resolving both smooth strength scaling and piecewise energy regimes.
- **3. Explainable Mechanics via TreeSHAP:** Confirmed physically consistent attributions: apparent density and 3D sphericity dominate crushing strength σ_f, while shortest axis S and equivalent diameter d_{eq} govern peak force F_f and crushing energy E_f.
- **4. Rapid Non-Destructive Virtual Suite:** Deployed an interactive 4-tab Streamlit engineering platform replacing slow, destructive laboratory trials with real-time screening.
- **5. Literature Validation & Engineering Screening:** Validated scaling accuracy within 1.15%–5.07% MAPE against Cai et al. (2024) and Liang et al. (2025), and integrated Li et al. (2021) fractal criteria with IRC:37-2018 / IS 383 aggregate screening.

7.2 Future Plan of Work

Building upon the outcomes of this study, future research directions include:

- **1. Indian C&D Waste Calibration:** Conduct laboratory UTM compression and mechanical sieving on locally sourced C&D waste from Indian recycling facilities (Delhi, Ahmedabad, Jodhpur) to establish regional calibration factors.
- **2. Discrete Element Method (DEM) Integration:** Couple the single-particle ML breakage engine into 3D DEM codes (PFC3D / YADE) using fragment-replacement algorithms to simulate bulk roller compaction and cyclic traffic loading.
- **3. 3D Laser / Structured-Light Scanner:** Upgrade the auxiliary computer vision module to multi-view 3D photogrammetry or structured-light scanning to directly capture the shortest axis S without 2D approximations.

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